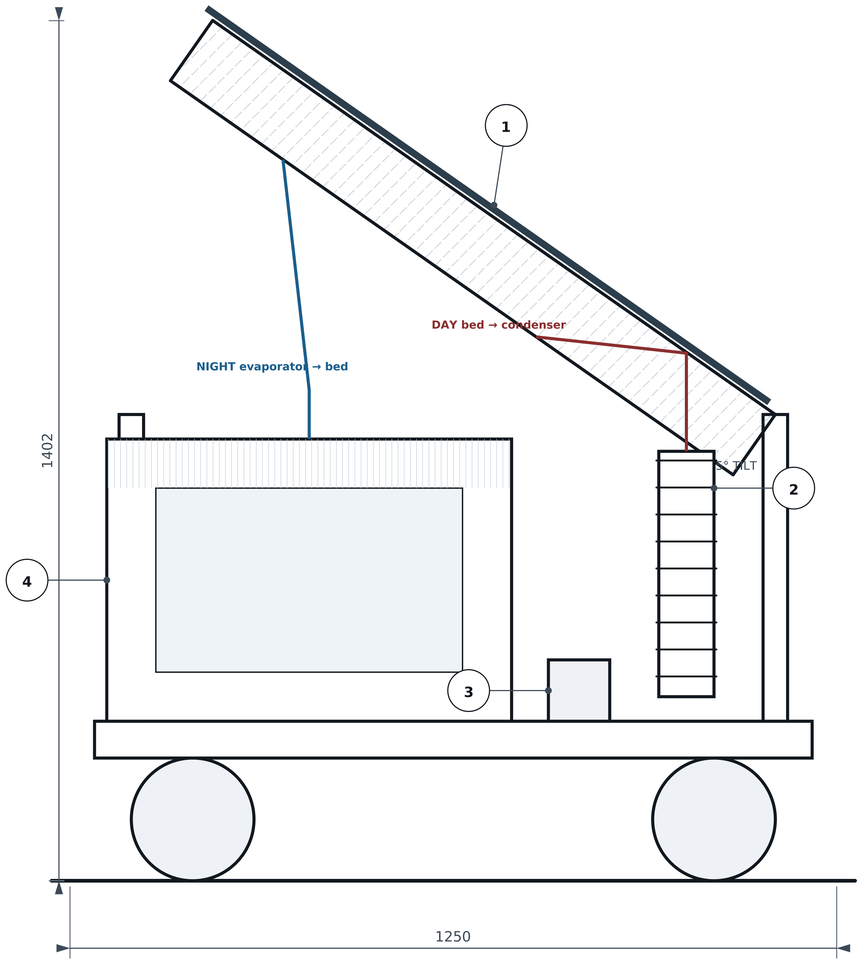
CONTAINS 4.9 kg OF METHANOL — TOXIC AND FLAMMABLE — OUTDOOR INSTALLATION ONLY, SEE SI-300

SIDE ELEVATION

SCALE 5:10

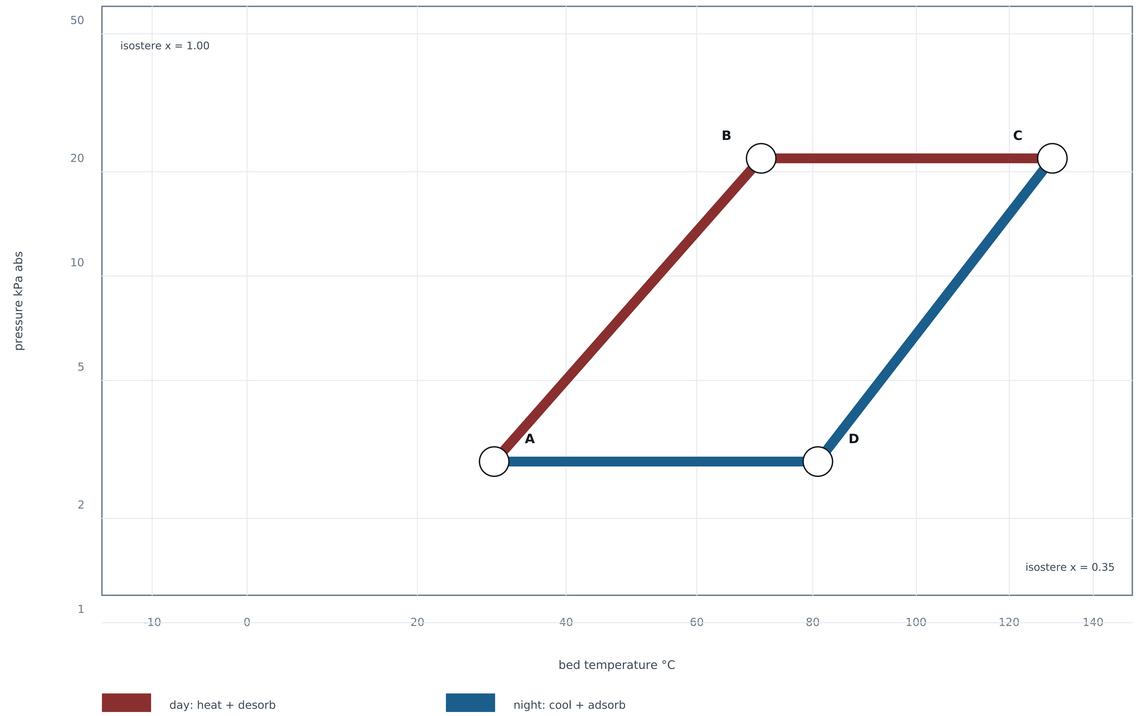


TOTAL MASS 91 kg CHARGED — a wheeled trolley, not a carried unit

MASS	
adsorber tubes, SS 316L	17.5 kg
absorber plate, Cu 0.5	4.5 kg
glazing, low-iron 3.2	8.0 kg
collector box + insulation	12.0 kg
activated carbon	4.9 kg
METHANOL CHARGE	4.9 kg
condenser + receiver	9.0 kg
evaporator + ice box	14.0 kg
frame, wheels, handle	16.0 kg
TOTAL, CHARGED	**90.8 kg**

CLAPEYRON CYCLE

ln P vs −1/T



BILL OF MATERIALS			
ITEM	PART	DESCRIPTION	QTY
1	SI-101	ADSORBER-COLLECTOR, 8 TUBE	1
2	SI-102	CONDENSER, FINNED, NATURAL CONVECTION	1
3	SI-103	RECEIVER, 3 L	1
4	SI-104	EVAPORATOR + ICE BOX	1
5	SI-105	CHECK VALVE, ALL-METAL	2
6	—	ACTIVATED CARBON, MAXSORB III	4.9 kg
7	—	METHANOL, ANHYDROUS >99.8 %	4.9 kg

INDIA-VC THERMAL HARDWARE — ISO 128 / ISO 2768-mK

TITLE
SOLAR ICE MAKER — ARRANGEMENT
ACTIVATED CARBON + METHANOL / 1.0 m2 / 5 kg ICE PER DAY / OUTDOOR ONLY

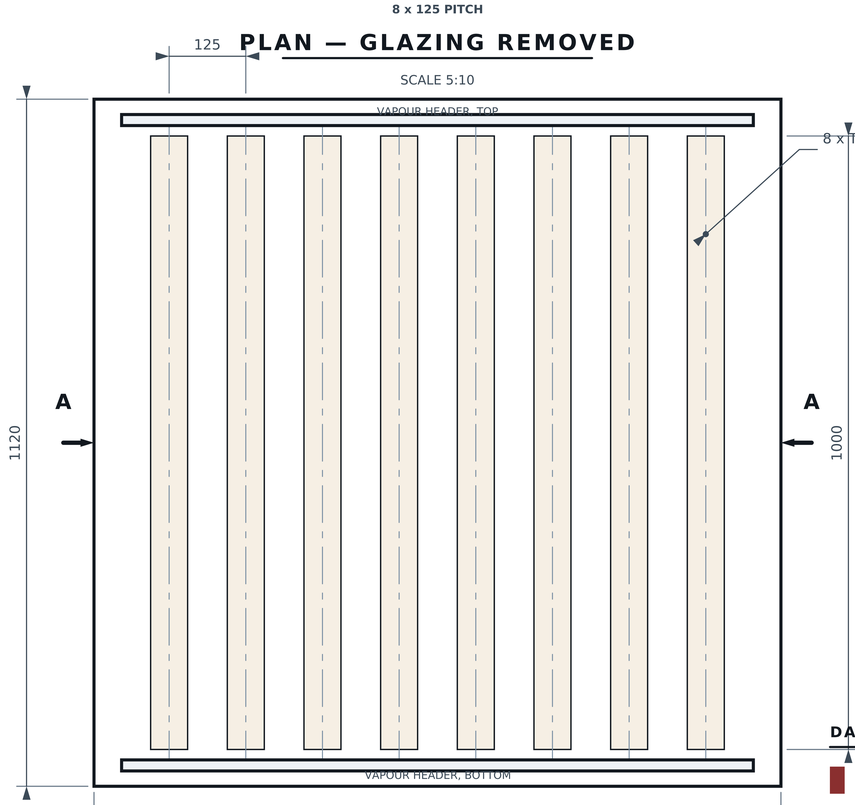
SCALE **5:10** — UNITS **mm** — THIRD ANGLE PROJECTION

DRAWING No. **SI-100** — REV **A** — SHEET **1 / 6**

DRAWN	CHECKED	MATERIAL	FINISH	DATE
AI-ASSISTED DRAFT	— PENDING —	SEE BOM	SEE NOTES	2026-09-15

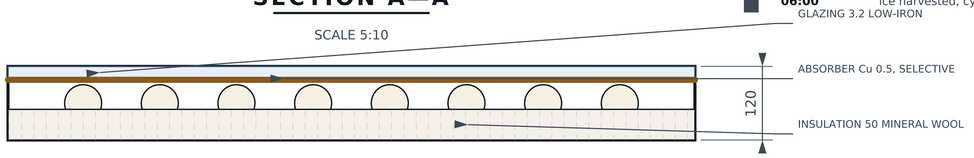
PLAN — GLAZING REMOVED

SCALE 5:10



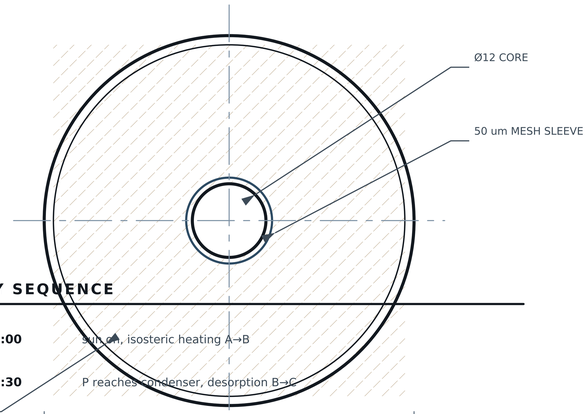
SECTION A—A

SCALE 5:10

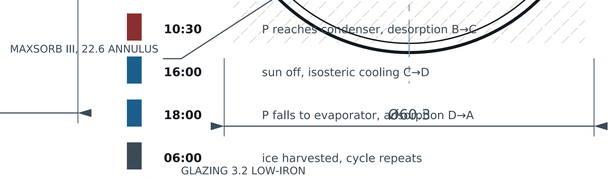


DETAIL B — TUBE SECTION

SCALE 5:1



DAILY SEQUENCE



TUBE DATA	
carbon per tube	0.61 kg
packed volume	16.4 L
fill of available	83 %
annulus	22.6 mm
diffusion time	11.8 min
operating pressure	8.4 bar
stagnation design	**14.0 bar**
hoop at stagnation	26.7 MPa
margin on SS 316L	**4.9 x**

NOTES

1. MATL: tubes and headers SS 316L. Absorber plate Cu, selective coating a 0.95 / e 0.10.
2. STAGNATION IS THE DESIGN CASE. A solar collector will stagnate; at 150 C methanol reaches 14.0 bar.
3. No pressure relief to atmosphere anywhere on this assembly. The tube contains stagnation instead.
4. All joints TIG welded, full penetration. No threaded or elastomer joint on the wetted side.
5. Bake carbon 200 C / 8 h under vacuum before filling; fill under dry nitrogen.
6. Core perforation 2.0 dia, 30 % open, sleeved in 50 um mesh to retain fines.

CRITICAL CHARACTERISTICS			
FEATURE	**NOMINAL**	**TOL**	**CRITICAL TO**
Tube wall	1.5	-0 / +0.15	stagnation containment
Weld, tube to header	full penetration	100 % dye pen	hermeticity
Absorber-to-tube bond	continuous	ultrasonic	collector efficiency
Carbon mass per tube	0.61 kg	±2 %	cycle capacity

INDIA-VC THERMAL HARDWARE — ISO 128 / ISO 2768-mK

TITLE
ADSORBER-COLLECTOR
SS 316L / 8 TUBES OD60.3 / 4.9 kg CARBON / DESIGN 14 bar

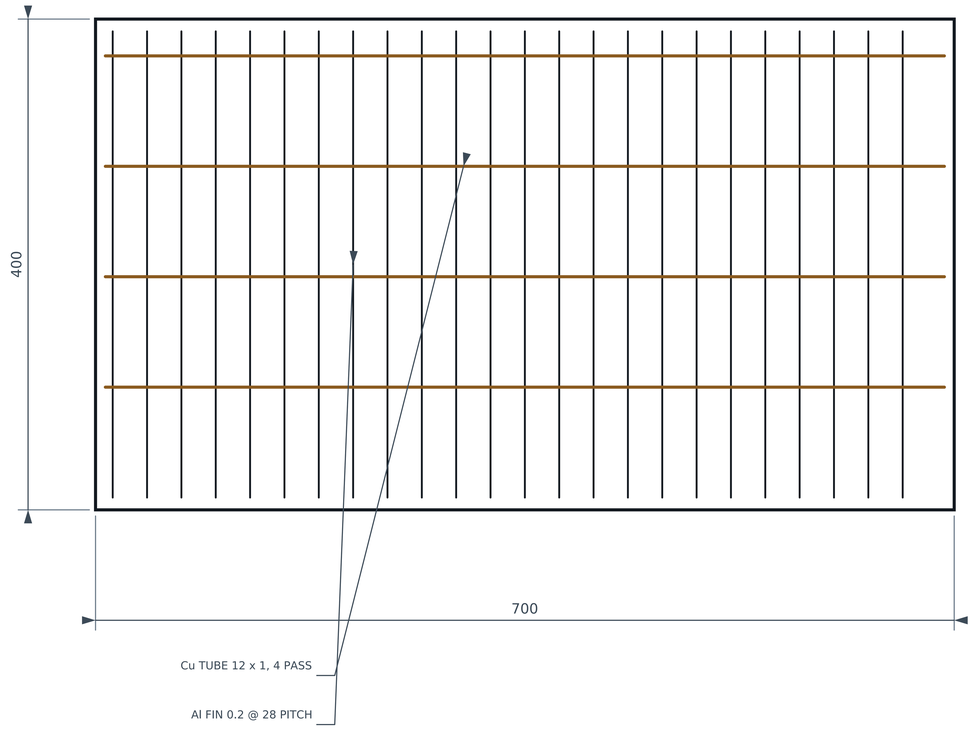
SCALE **5:10** — THIRD ANGLE PROJECTION
UNITS **mm**

DRAWING No. **SI-101**
REV **A**
SHEET **2 / 6**

DRAWN	CHECKED	MATERIAL	FINISH	DATE
AI-ASSISTED DRAFT	— PENDING —	SEE BOM	SEE NOTES	2026-09-15

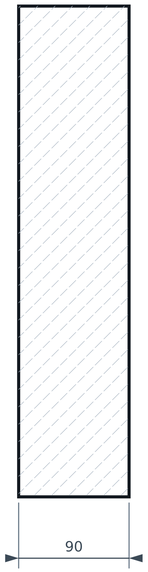
CONDENSER — FRONT

SCALE 1:5

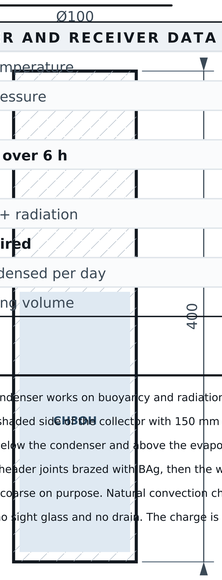


SIDE

1:5



RECEIVER SI-103



CONDENSER AND RECEIVER DATA	
condensing temperature	30 C
condensing pressure	21.9 kPa abs
duty per day	3.76 MJ
mean power over 6 h	**174 W**
air-side dT	15 K
h, convection + radiation	10 W/m2K
surface required	**1.16 m2**
methanol condensed per day	3.19 kg
receiver working volume	3.0 L

NOTES

1. No fan. The condenser works on buoyancy and radiation, so it must be shaded from the sun and free to draw air from below.
2. Mount on the shaded side of the collector with 150 mm clear beneath and above.
3. Receiver sits below the condenser and above the evaporator: the loop is gravity-driven, there is no pump.
4. Cu tube to SS header joints brazed with BAg, then the whole assembly TIG welded into the envelope.
5. Fin pitch 28 is coarse on purpose. Natural convection chokes below about 6 mm free spacing.
6. Receiver has no sight glass and no drain. The charge is not serviceable in the field.

WORKING LEVEL 3.2 kg

INSPECTION		
CHECK	**REQUIREMENT**	**METHOD**
Helium leak, whole assembly	< 1e-8 Pa.m3/s	mass spectrometer
Proof pressure	21 bar, 5 min	hydrostatic, then dry
Condenser orientation	tubes horizontal, drain to receiver	level check
Shading	no direct sun on the fins	site survey

INDIA-VC THERMAL HARDWARE — ISO 128 / ISO 2768-mK

TITLE: **CONDENSER AND RECEIVER**
AIR-COOLED, NATURAL CONVECTION / 174 W MEAN / 1.16 m2

SCALE: **1:5** — THIRD ANGLE PROJECTION
UNITS: **mm**

DRAWING No.: **SI-102 / SI-103**
REV: **A**
SHEET: **3 / 6**

DRAWN	CHECKED	MATERIAL	FINISH	DATE
AI-ASSISTED DRAFT	— PENDING —	SEE BOM	SEE NOTES	2026-09-15

ICE BOX — SECTION

SCALE 1:5

EVAPORATOR AND BOX DATA	
evaporating temperature	-5 C
evaporating pressure	2.92 kPa abs
cooling delivered per night	2.19 MJ
mean power over 12 h	**51 W**
surface required	**508 cm2**
box internal	500 x 350 x 300
insulation U	0.30 W/m2K
heat leak at 30 C ambient	8 W
leak as share of the night	**15 %**

NOTES

1. The evaporator runs at 2.9 kPa absolute. It sees full external atmospheric pressure whenever the machine is cold - design it as a vacuum vessel, not a pressure vessel.
2. Plate-coil in SS 316L, all welded, no fittings inside the box.
3. Evaporator sits ABOVE the trays so liquid methanol drains back to the receiver by gravity when the machine warms.
4. Secondary containment: the box is lined with a welded SS tray connected to the bund on SI-300. A leak must not reach the ice.
5. The ice is NOT potable. Methanol is on the other side of one wall. Use the ice for cooling, not for drinks.
6. 80 mm PU gives a 15 % overnight leak, already inside the 5 kg figure.

INSPECTION		
CHECK	**REQUIREMENT**	**METHOD**
Evaporator external collapse	full vacuum, no set	1 bar external, 1 h
Helium leak	< 1e-8 Pa.m3/s	mass spectrometer
Containment tray	welded, drains to bund	water fill test
Box heat leak	<= 10 W at 30 C	calibrated pull-down

INDIA-VC THERMAL HARDWARE — ISO 128 / ISO 2768-mK

TITLE: **EVAPORATOR AND ICE BOX**
5 kg ICE PER NIGHT / 51 W MEAN / 80 mm PU

SCALE: **1:5** — THIRD ANGLE PROJECTION
UNITS: **mm**

DRAWING No. **SI-104**
REV **A**
SHEET **4 / 6**

DRAWN	CHECKED	MATERIAL	FINISH	DATE
AI-ASSISTED DRAFT	— PENDING —	SEE BOM	SEE NOTES	2026-09-15

CHARGING IS THE MOST DANGEROUS OPERATION IN THE LIFE OF THIS MACHINE — OUTDOORS, TRAINED OPERATOR, NO IGNITION SOURCE

CHARGING SEQUENCE

1 — WELD OUT
Complete every wetted joint. No threaded or elastomer joint remains.

2 — PROOF PRESSURE — 1.5 x stagnation
Hydrostatic 21 bar, 5 min. Drain, dry with warm dry nitrogen.

3 — LEAK TEST — < 1e-8 Pa.m3/s
Helium mass spectrometer under vacuum on the whole envelope.

4 — BAKE-OUT — carbon outgasses
200 C / 8 h at < 1 Pa through the charge tube, pumping throughout.

5 — COOL UNDER VACUUM — no air, ever
Let the bed reach ambient still under the pump. Do not vent.

6 — CHARGE — by mass, not volume
Admit 4.9 kg anhydrous methanol by WEIGHT from a sealed burette.

7 — DEGAS — removes dissolved air
Freeze-pump-thaw x3 on the receiver. Final pump to < 50 Pa at 0 C.

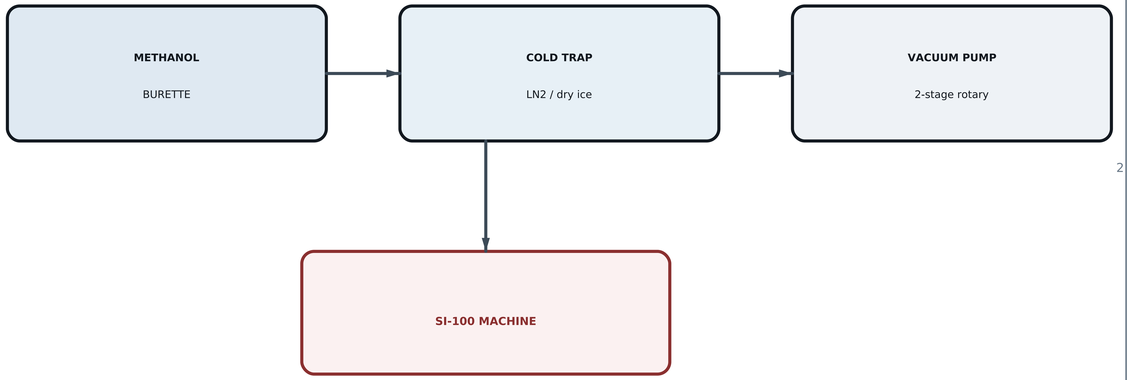
8 — PINCH-OFF — permanent
Cold-weld the charge tube, cut, then TIG cap the stub.

9 — LEAK TEST 2 — < 1e-8 Pa.m3/s
Helium bomb 3 bar / 4 h, then mass spectrometer.

10 — FIRST CYCLE — commissioning
One full solar cycle under observation. Record bed and receiver temperatures.

CHARGING RIG

SCHEMATIC



The cold trap protects the pump and, more importantly, stops methanol

vapour reaching the exhaust. Vent the pump outdoors, downwind.

CHARGING PARAMETERS

PARAMETER	VALUE	LIMIT
Methanol charge	**4.91 kg**	**by weight, ±1 %**
Purity	anhydrous > 99.8 %	water poisons the carbon
Bake-out	200 C / 8 h	< 1 Pa
Final vacuum before seal	< 50 Pa	at 0 C
Proof pressure	21 bar	1.5 x stagnation
Leak rate	**< 1e-8 Pa.m3/s**	**helium, twice**
Seal	cold-weld + TIG cap	no valve, no port

PROHIBITED

- Venting the envelope to air at any point after bake-out. Air in the bed is non-condensable gas and it never comes out again.
- Charging by volume. Methanol expands 1.2 % per 10 K; the charge is a mass specification.
- Any valve, gauge port or sight glass left on the wetted envelope. Every one is a leak path for the life of the machine.
- Charging indoors, or anywhere without free ventilation and no ignition source within 5 m.

INDIA-VC THERMAL HARDWARE				ISO 128 / ISO 2768-mK
TITLE: VACUUM AND CHARGING 4.9 kg METHANOL / ONE-TIME PERMANENT CHARGE / NO FIELD SERVICE			SCALE: NONE THIRD ANGLE PROJECTION	UNITS: mm
DRAWING No.: SI-200			REV: A	SHEET: 5 / 6
DRAWN: AI-ASSISTED DRAFT	CHECKED: — PENDING —	MATERIAL: SEE BOM	FINISH: SEE NOTES	DATE: 2026-09-15

THIS IS NOT A CERTIFIABLE DOMESTIC APPLIANCE

Methanol has no ASHRAE refrigerant designation for this service and grades as both toxic and highly flammable. There is no path to IEC 60335-2-40 approval for indoor use.

HAZARD DATA		
PROPERTY	**VALUE**	**CONSEQUENCE**
Charge	**4.9 kg methanol**	**outdoor only**
Flash point	11 C closed cup	below ambient
Flammable range	6 – 36 % in air	unusually wide
Flame	nearly invisible in daylight	cannot be seen
Acute toxicity	10 mL blinds, 30 mL can kill	skin absorbs it
Occupational limit	200 ppm TWA	low
Operating pressure	8.4 bar at 130 C	vessel, not plumbing
Stagnation pressure	**14.0 bar at 150 C**	**the design case**

DESIGN CONSEQUENCES ALREADY BUILT IN

- All-welded SS 316L and copper envelope. No elastomer, no thread, no service port anywhere wetted.
- One-time permanent charge, sealed by cold-weld pinch-off and a welded cap. The machine is replaced, never opened.
- No pressure relief to atmosphere. The tubular adsorber contains stagnation at 4.9 x margin instead of venting methanol.
- Welded stainless containment tray under the evaporator, receiver and condenser, drained to a bund sized for the whole charge.
- Ice box lined and sealed: a leak cannot reach the ice. The ice is not potable in any case.

MACHINE SUMMARY	
Designed output	5 kg ice / day
Solar COP	0.110
Collector	1.0 m2
Total mass, charged	91 kg
Certification status	**NONE — demonstrator**

REQUIRED MARKING

ON THE VESSEL AND THE FRAME

METHANOL CH3OH

4.9 kg — TOXIC, FLAMMABLE

INVISIBLE FLAME

OUTDOOR USE ONLY · DO NOT OPEN · DO NOT VENT

ICE NOT POTABLE · NO IGNITION SOURCE WITHIN 5 m

COMMISSIONING		
STEP	**COMMISSIONING**	**RECORD**
1	Site: outdoors, level, free air, 5 m from ignition	site sheet
2	Confirm bund and tray drain clear	visual
3	Leak test 2 witnessed before first sun	He, < 1e-8
4	First solar cycle observed end to end	T bed, T receiver
5	Confirm condenser drains fully to the receiver	no hold-up
6	Measure ice after night one	kg, vs 5 kg target
7	Record stagnation temperature on a no-load day	T max

IF THE MACHINE IS DAMAGED

- Do not approach a suspected leak with any ignition source. The flame is invisible in daylight; use a thermal camera or a broom handle.
- Methanol vapour is heavier than air near the source and pools in the bund. Ventilate from upwind.
- Do not attempt to recover the charge. Isolate the machine, let it weather in open air, and dispose of the carbon and vessel through a chemical waste route.
- A machine that has been vented to atmosphere is finished as a machine: air in the bed cannot be pumped out in the field.

INDIA-VC THERMAL HARDWARE				ISO 128 / ISO 2768-mK
TITLE: **SAFETY, MARKING AND COMMISSIONING** OUTDOOR OFF-GRID EQUIPMENT / 4.9 kg METHANOL / NOT A DOMESTIC APPLIANCE			SCALE: **NONE** THIRD ANGLE PROJECTION	UNITS: **mm**
DRAWING No.: **SI-300**			REV: **A**	SHEET: **6 / 6**
DRAWN: AI-ASSISTED DRAFT	CHECKED: — PENDING —	MATERIAL: SEE BOM	FINISH: SEE NOTES	DATE: 2026-09-15